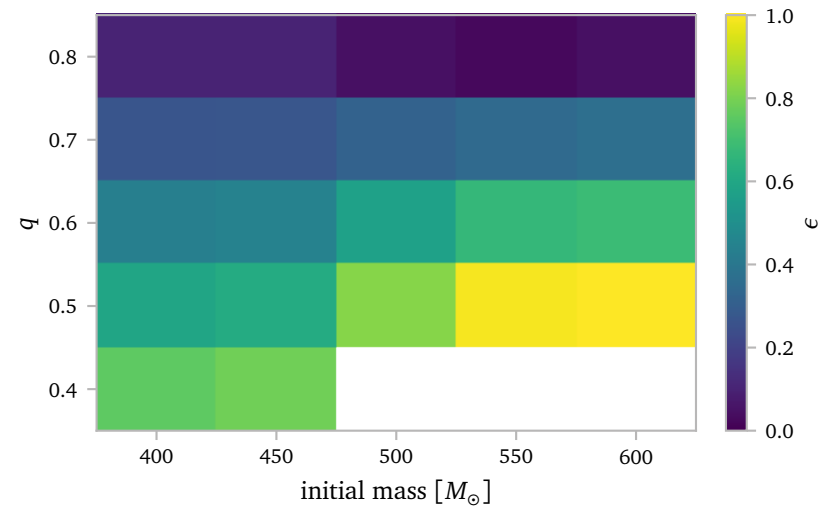


Notes Week 23

a. Grid

Here I will note down my most important findings for the grid that is based on the accretion efficiency required to produce a $1.29 M_{\odot}$ barium star.

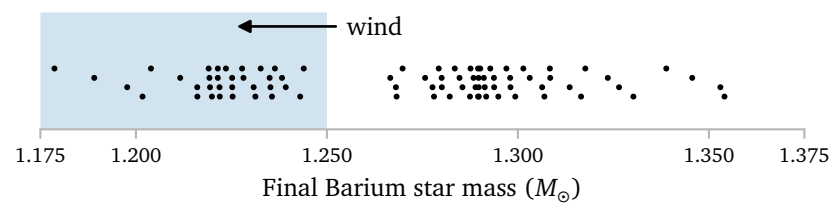


This plot show the ϵ values for the different $R_{\text{Roche lobe}}$ and q combinations. Note that the bottom right 3 squares are empty because they cannot produce a $1.29 M_{\odot}$ barium star, even with 100% accretion. The grid only goes up to $q = 0.8$ because $q = 0.9$ would produce barium stars heavier than $1.29 M_{\odot}$ even without any accretion.

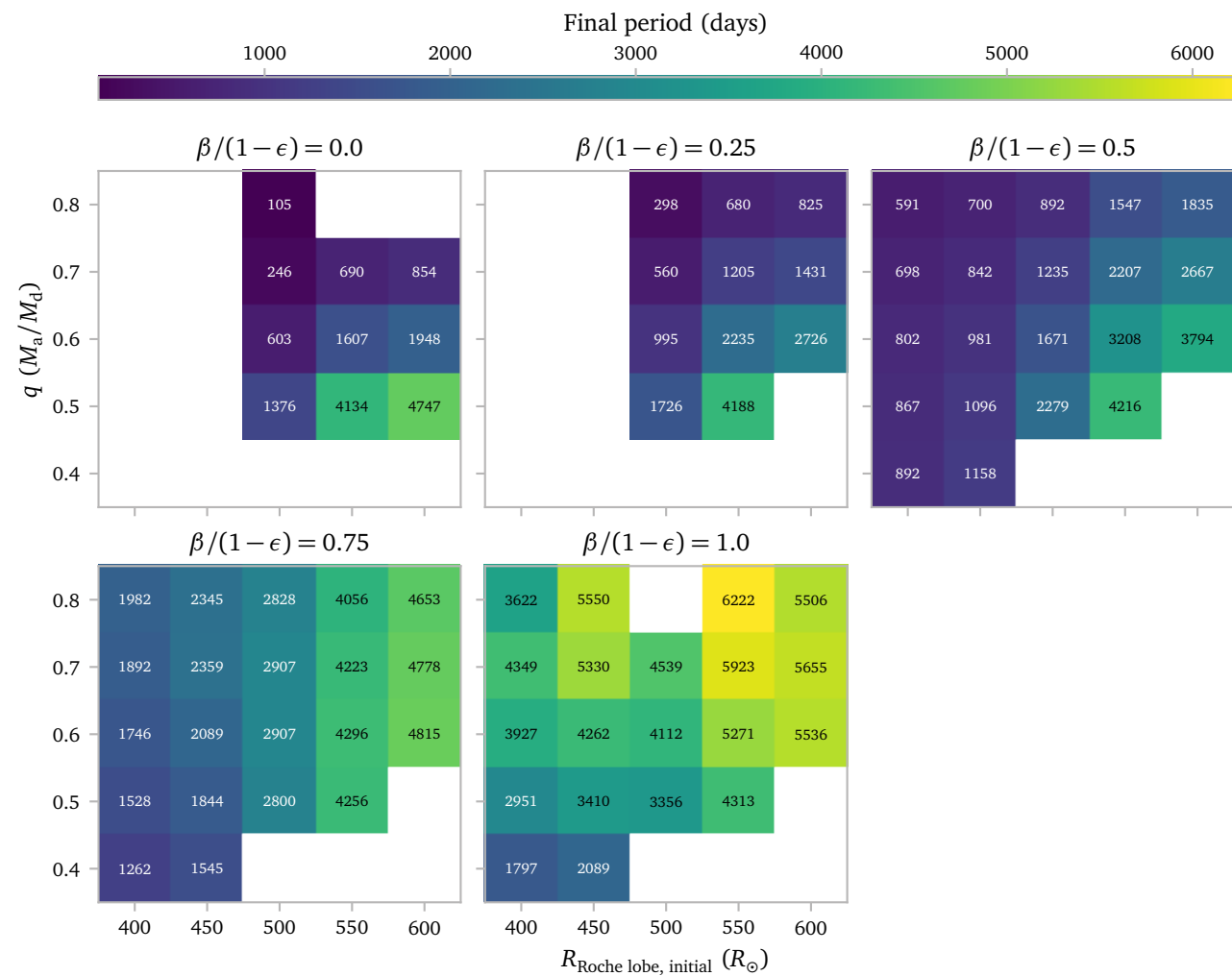
Below I investigate the 3d¹ grid that I simulated last week.

The first thing I want to check is *how* close the final accretor masses are to the $1.29 M_{\odot}$ value.

If there would be no wind accretion after the initialization of the models, and they all interact at the predicted time, all models should have a mass of $1.29 M_{\odot}$.

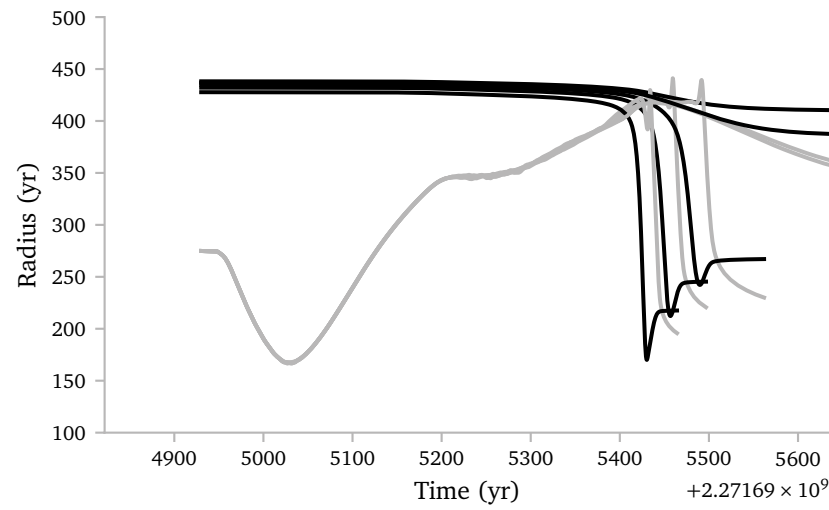


1 5 × 5 × 5



b. fix

Some of the models in the grid interact directly after starting.



This is not *necessarily* wrong but it is *better* to start the simulation a bit earlier, so we know for sure the stars would not have interacted at an earlier pulse.

With the refactor of the old grid-generation code, I accidentally forgot to take the model BEFORE the first model that has a radius smaller than $0.9 \times R_{\text{RL}}$. Below, I show the *new* and *corrected* version that does have this feature again.

```
def select_model(contact_age, models_dict):  
    candidate = None  
    models = list(models_dict.values())  
  
    for i in range(len(models) - 1, -1, -1):  
        if models[i]["age"] < contact_age:  
            # Return the previous model if it exists  
            if i > 0:  
                return models[i - 1]  
  
    print("no suitable stellar model before contact ->  
    ~ skipping")  
    return candidate
```

References